

现代控制理论 · 第二周作业标准解答

习题七：7-9, 7-10(a)(c), 7-12(a), 7-15, 7-17

7-9

已知

$$G(z) = \frac{Y(z)}{U(z)} = \frac{z+1}{z^2-1.4z+0.48},$$

输入为单位阶跃, 故

$$U(z) = \frac{z}{z-1}.$$

系统极点满足

$$z^2 - 1.4z + 0.48 = (z - 0.8)(z - 0.6),$$

均位于单位圆内, 因此可用离散终值定理:

$$\begin{aligned} y(\infty) &= \lim_{z \rightarrow 1} \frac{z-1}{z} Y(z) \\ &= \lim_{z \rightarrow 1} \frac{z-1}{z} G(z) \frac{z}{z-1} = G(1) \\ &= \frac{1+1}{1-1.4+0.48} = 25. \end{aligned}$$

$$\boxed{y(\infty) = 25}$$

7-10(a)

记连续环节 $F(s)$ 的脉冲传递函数为

$$F(z) = \mathcal{Z} \{ \mathcal{L}^{-1}[F(s)]|_{t=kT} \}.$$

设主采样器输出为 $E(z)$ 。由前向通道

$$Y(z) = H(z)E(z).$$

反馈支路中, $H(s)$ 与 $G_2(s)$ 之间没有采样器, 因此二者应先在 s 域相乘后再作 z 变换, 记

$$HG_2(z) = \mathcal{Z}\{H(s)G_2(s)\},$$

则

$$V(z) = HG_2(z)E(z).$$

经过反馈支路采样器后, $G_1(s)$ 与前级之间被采样器隔开, 故

$$B(z) = G_1(z)V(z) = G_1(z)HG_2(z)E(z).$$

主采样点满足

$$E(z) = R(z) - B(z),$$

故

$$E(z)[1 + G_1(z)HG_2(z)] = R(z).$$

于是闭环脉冲传递函数

$$\boxed{\frac{Y(z)}{R(z)} = \frac{H(z)}{1 + G_1(z)HG_2(z)}}.$$

7-10(c)

定义

$$G_{01}(z) = \mathcal{Z}\{H_0(s)G_1(s)\}, \quad G_{012}(z) = \mathcal{Z}\{H_0(s)G_1(s)G_2(s)\}.$$

设数字控制器输入为 $E(z)$, 则

$$D(z) = G_D(z)E(z).$$

由结构得

$$C(z) = G_{01}(z)D(z), \quad Y(z) = G_{012}(z)D(z).$$

两个反馈均为负反馈, 故

$$E(z) = X(z) - C(z) - Y(z).$$

代入可得

$$E(z) = X(z) - G_D(z)G_{01}(z)E(z) - G_D(z)G_{012}(z)E(z),$$

即

$$E(z) = \frac{X(z)}{1 + G_D(z)G_{01}(z) + G_D(z)G_{012}(z)}.$$

所以

$$\boxed{\frac{Y(z)}{X(z)} = \frac{G_D(z)G_{012}(z)}{1 + G_D(z)G_{01}(z) + G_D(z)G_{012}(z)}}.$$

7-12(a)

记

$$G_1(z) = \mathcal{Z}\{G_1(s)\}, \quad G_3(z) = \mathcal{Z}\{G_3(s)\}, \quad G_{12}(z) = \mathcal{Z}\{G_1(s)G_2(s)\}.$$

由于 $G_1(s)$ 与 $G_2(s)$ 之间没有采样器, 内反馈通道应使用整体脉冲传递函数 $G_{12}(z)$ 。

设第一个采样器输出为 $A(z)$, 第二个加法点送入 $G_1(s)$ 的采样序列为 $Q(z)$ 。内环满足

$$Q(z) = A(z) - G_{12}(z)Q(z),$$

故

$$A(z) = [1 + G_{12}(z)]Q(z).$$

输出采样值为

$$C(z) = G_1(z)Q(z).$$

外反馈支路满足

$$A(z) = R(z) - G_3(z)C(z) = R(z) - G_3(z)G_1(z)Q(z).$$

联立得

$$[1 + G_{12}(z) + G_1(z)G_3(z)]Q(z) = R(z).$$

因此

$$\boxed{\frac{C(z)}{R(z)} = \frac{G_1(z)}{1 + G_{12}(z) + G_1(z)G_3(z)}}.$$

7-15

连续状态方程

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t) + Bu(t), \quad y(t) = C\mathbf{x}(t),$$

其中

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

采样周期为 T ，零阶保持输入下

$$\mathbf{x}(k+1) = \Phi\mathbf{x}(k) + \Gamma u(k),$$

其中

$$\Phi = e^{AT}, \quad \Gamma = \int_0^T e^{A\tau} B d\tau.$$

因为 $A^2 = 0$,

$$e^{A\tau} = I + A\tau = \begin{bmatrix} 1 & \tau \\ 0 & 1 \end{bmatrix},$$

故

$$\Phi = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix},$$

并且

$$\Gamma = \int_0^T \begin{bmatrix} 1 & \tau \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} d\tau = \int_0^T \begin{bmatrix} \tau \\ 1 \end{bmatrix} d\tau = \begin{bmatrix} T^2/2 \\ T \end{bmatrix}.$$

因此离散状态方程为

$$\boxed{\mathbf{x}(k+1) = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} T^2/2 \\ T \end{bmatrix} u(k)}, \quad \boxed{y(k) = \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x}(k)}.$$

7-17

已知

$$\mathbf{x}(k+1) = A\mathbf{x}(k) + Bu(k), \quad y(k) = C\mathbf{x}(k),$$

其中

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0.$$

零初始条件下

$$G(z) = C(zI - A)^{-1}B + D.$$

有

$$zI - A = \begin{bmatrix} z-1 & -1 \\ 0 & z-1 \end{bmatrix},$$

所以

$$(zI - A)^{-1} = \begin{bmatrix} \frac{1}{z-1} & \frac{1}{(z-1)^2} \\ 0 & \frac{1}{z-1} \end{bmatrix}.$$

于是

$$\begin{aligned} G(z) &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{z-1} & \frac{1}{(z-1)^2} \\ 0 & \frac{1}{z-1} \end{bmatrix} \begin{bmatrix} 1/2 \\ 1 \end{bmatrix} \\ &= \frac{1}{2(z-1)} + \frac{1}{(z-1)^2} = \frac{z+1}{2(z-1)^2}. \end{aligned}$$

因此

$$G(z) = \frac{Y(z)}{U(z)} = \frac{z+1}{2(z-1)^2} = \frac{0.5z+0.5}{z^2-2z+1}.$$